Assignment 3
Uncertainty and Bayesian Networks

Hashim AlSayed · Student ID S24109290
Effat University · College of Engineering · CS3081 Artificial Intelligence
Instructor: Dr. Naila Marir

Worked solutions to Exercises 1–4 and 5(a),(b), with a summary of 5(c). Final probabilities are rounded to four decimal places. Exact fractions are retained wherever necessary to prevent intermediate rounding errors.

Exercise 1 — Joint distribution

Weather	On time	Late	Weather total
Clear	0.5600	0.1400	0.7000
Dusty	0.1200	0.1800	0.3000
Shuttle total	0.6800	0.3200	1.0000

1. Validity

Every entry is nonnegative and at most 1. Their sum is 0.5600 + 0.1400 + 0.1200 + 0.1800 = 1.0000, so this is a valid joint probability distribution.

2. Marginal probabilities

Sum over the other variable:
P(dusty) = P(dusty, on time) + P(dusty, late) = 0.1200 + 0.1800 = **0.3000**.
P(late) = P(clear, late) + P(dusty, late) = 0.1400 + 0.1800 = **0.3200**.

3. Conditional probabilities

P(late | dusty) = P(late, dusty) / P(dusty) = 0.1800 / 0.3000 = **0.6000**.
P(clear) = 0.5600 + 0.1400 = 0.7000.
P(late | clear) = 0.1400 / 0.7000 = **0.2000**.

4. Reverse conditional

P(dusty | late) = 0.1800 / 0.3200 = **0.5625**.
This is not 0.6000 because reversing the condition changes the reference population and denominator.

5. Independence

Independence would require P(dusty, late) = P(dusty)P(late), but 0.1800 ≠ 0.3000 × 0.3200 = 0.0960; therefore weather and shuttle status are **not independent**.

Exercise 2 — Bayes' rule: a spam filter

Let S = spam, N = normal, F = contains "free", and W = contains "winner". S and N are mutually exclusive and exhaustive.

1. Probability that an email contains "free"

Use the **law of total probability**, splitting all emails into spam and normal:
$P(F) = P(F \mid S)P(S) + P(F \mid N)P(N)$
$= (0.6000 \times 0.3000) + (0.0500 \times 0.7000)$
$= 0.1800 + 0.0350 = \mathbf{0.2150}$.

2. Spam given "free"

Use **Bayes' rule**:
$P(S \mid F) = P(F \mid S)P(S) / P(F)$
$= (0.6000 \times 0.3000) / 0.2150$
$= 0.1800 / 0.2150 = 36/43$.

P(spam | free) = 0.8372

3. Spam given both words

By the stated conditional independence assumption:
$P(F, W \mid S) = P(F \mid S)P(W \mid S) = 0.6000 \times 0.4000 = 0.2400$.
$P(F, W \mid N) = P(F \mid N)P(W \mid N) = 0.0500 \times 0.0200 = 0.0010$.

Compute the two unnormalized contributions:
$P(S, F, W) = 0.3000 \times 0.2400 = 0.0720$.
$P(N, F, W) = 0.7000 \times 0.0010 = 0.0007$.
$P(F, W) = 0.0720 + 0.0007 = 0.0727$.

Normalize using Bayes' rule:
$P(S \mid F, W) = 0.0720 / 0.0727 = 720/727$.

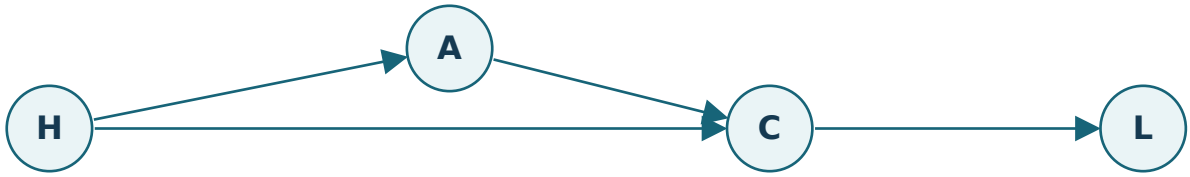
P(spam | free, winner) = 0.9904

4. Comparison (one sentence)

The spam probability increases from **0.8372** to **0.9904** because "winner" is much more likely in spam than normal email (0.4000 versus 0.0200), adding evidence in favor of spam.

Exercise 3 — Structure and joint probability

1. Network and parents



Node	Meaning	Parents
H	Heatwave	None
A	AC overload	H
C	Power cut	H, A
L	Lab cancelled	C

2. Joint factorization

P(H, A, C, L) = P(H) P(A | H) P(C | H, A) P(L | C)

3. Number of independent probabilities

Each binary node needs one “yes” number per parent configuration:
H: $2^0 = 1$; A: $2^1 = 2$; C: $2^2 = 4$; L: $2^1 = 2$.
Total = 1 + 2 + 4 + 2 = **9 numbers**, rather than $2^4 - 1 = 15$ for an unrestricted joint table; each “no” entry is determined by its complement.

4. Specified joint event

P(H=yes, A=no, C=yes, L=yes)
= P(H=yes) P(A=no | H=yes) P(C=yes | H=yes,A=no) P(L=yes | C=yes)
= 0.3000 × (1 − 0.6000) × 0.2000 × 0.8000
= 0.3000 × 0.4000 × 0.2000 × 0.8000 = **0.0192**.

5. Heatwave given a power cut

First sum over A:
P(C=yes | H=yes) = 0.6000×0.7000 + 0.4000×0.2000 = 0.5000.
P(C=yes | H=no) = 0.1000×0.5000 + 0.9000×0.0500 = 0.0950.
P(C=yes) = 0.3000×0.5000 + 0.7000×0.0950 = 0.2165.
P(H=yes | C=yes) = (0.3000×0.5000) / 0.2165 = 300/433 = **0.6928**.

Also **P(H=yes | C=yes,L=yes) = 0.6928**, because L is conditionally independent of H given its only parent C; after C is observed, L adds no information about H.

Exercise 4 — Enumeration

1. Find P(H=yes | L=yes)

Query: H. **Evidence:** L=yes. **Hidden variables:** A and C.
For each h, define w(h) = P(H=h,L=yes):
$w(h) = P(h) \Sigma_{a \in \{yes,no\}} P(a \mid h) \Sigma_{c \in \{yes,no\}} P(c \mid h,a) P(L=yes \mid c)$.

The following eight terms explicitly enumerate every combination of H, A and C. Fractions in the last column are exact; probability displays are rounded to four places.

H	A	C	Product (L=yes)	Exact weight
yes	yes	yes	0.3000×0.6000×0.7000×0.8000	2016/20000
yes	yes	no	0.3000×0.6000×0.3000×0.1000	108/20000
yes	no	yes	0.3000×0.4000×0.2000×0.8000	384/20000
yes	no	no	0.3000×0.4000×0.8000×0.1000	192/20000
no	yes	yes	0.7000×0.1000×0.5000×0.8000	560/20000
no	yes	no	0.7000×0.1000×0.5000×0.1000	70/20000
no	no	yes	0.7000×0.9000×0.0500×0.8000	504/20000
no	no	no	0.7000×0.9000×0.9500×0.1000	1197/20000

Explicitly sum the hidden variables for each H value:
w(yes) = 0.3000{0.6000[0.7000×0.8000 + 0.3000×0.1000]
+ 0.4000[0.2000×0.8000 + 0.8000×0.1000]}
= 0.3000[0.6000×0.5900 + 0.4000×0.2400]
= 0.3000×0.4500 = 2700/20000 = **0.1350**.

w(no) = 0.7000{0.1000[0.5000×0.8000 + 0.5000×0.1000]
+ 0.9000[0.0500×0.8000 + 0.9500×0.1000]}
= 0.7000[0.1000×0.4500 + 0.9000×0.1350]
= 2331/20000, reported as **0.1166**.

Normalize without rounding the weights:
Z = P(L=yes) = (2700+2331)/20000 = 5031/20000 (reported as 0.2516).
Normalization factor α = 1/Z = 20000/5031.
P(H=yes | L=yes) = αw(yes) = 2700/5031 = 900/1677.
P(H=no | L=yes) = αw(no) = 2331/5031 = **0.4633**.

P(H=yes | L=yes) = 0.5367

Exercise 4 — Further evidence

2. Find P(A=yes | L=yes)

Query: A. **Evidence:** L=yes. **Hidden variables:** H and C.
For each a, $w(a) = \Sigma_h P(h)P(a \mid h) \Sigma_c P(c \mid h,a)P(L=yes \mid c)$.

For A=yes:
w(A=yes) = 0.3000×0.6000[0.7000×0.8000 + 0.3000×0.1000]
+ 0.7000×0.1000[0.5000×0.8000 + 0.5000×0.1000]
= 0.1800×0.5900 + 0.0700×0.4500
= 0.1062 + 0.0315 = 2754/20000 = **0.1377**.

For A=no:
w(A=no) = 0.3000×0.4000[0.2000×0.8000 + 0.8000×0.1000]
+ 0.7000×0.9000[0.0500×0.8000 + 0.9500×0.1000]
= 0.1200×0.2400 + 0.6300×0.1350
= 576/20000 + 1701/20000 = 2277/20000 (reported as 0.1139).

Z = (2754 + 2277)/20000 = 5031/20000, the same evidence probability as part 1.
P(A=yes | L=yes) = (2754/20000)/(5031/20000) = 918/1677.
P(A=no | L=yes) = 2277/5031 = **0.4526**.

P(A=yes | L=yes) = 0.5474

3. Find P(H=yes | L=yes, A=no)

Query: H. **Evidence:** L=yes and A=no. **Hidden variable:** C.
$w(h) = P(h)P(A=no \mid h) \Sigma_c P(c \mid h,A=no)P(L=yes \mid c)$.

w(H=yes) = 0.3000×0.4000[0.2000×0.8000 + 0.8000×0.1000]
= 0.1200×0.2400 = 576/20000 = **0.0288**.
w(H=no) = 0.7000×0.9000[0.0500×0.8000 + 0.9500×0.1000]
= 0.6300×0.1350 = 1701/20000 (reported as 0.0851).

Z = (576+1701)/20000 = 2277/20000; α = 20000/2277.
P(H=yes | L=yes,A=no) = 576/2277 = 192/759.
The complementary probability is 1701/2277 = **0.7470**.

P(H=yes | L=yes, A=no) = 0.2530 (down from 0.5367)

No overload is evidence against a heatwave because P(A=no | H=yes)=0.4000 is lower than P(A=no | H=no)=0.9000. Although a heatwave can still cause a power cut directly through H → C → L, observing A=no removes the overload-mediated route H → A → C → L. In this network with these tables, the posterior therefore decreases from 0.5367 to 0.2530.

Exercise 5 — Approximate inference

(a) Rejection sampling

1. Retain only samples with L=yes.
Keep: **2, 4, 6, 7, 12** (5 samples).
Throw away: **1, 3, 5, 8, 9, 10, 11** (7 samples).
Among the retained rows, H=yes occurs in samples 2 and 7, so the numerator is 2 and the denominator is 5.

Estimated P(H=yes | L=yes) = 2/5 = 0.4000

2. Rejected fraction: 7/12 = **0.5833**.
Rejected samples do not contribute to the posterior estimate, wasting computation; when the evidence is rare, very few generated samples survive, increasing the sample requirement and estimator variability.

(b) Likelihood weighting

1. Fix L=yes in every row. The weight is P(L=yes | C), so use 0.8000 when C=yes and 0.1000 when C=no; do not multiply in H, A, or C probabilities again.

Sample	H	C	Weight P(L=yes | C)
1	no	no	0.1000
2	yes	yes	0.8000
3	no	no	0.1000
4	yes	no	0.1000
5	no	yes	0.8000
6	yes	yes	0.8000

2. H=yes rows are 2, 4 and 6.
Their total weight is 0.8000 + 0.1000 + 0.8000 = **1.7000**.
All weights sum to 0.1000+0.8000+0.1000+0.1000+0.8000+0.8000 = **2.7000**.
Weighted estimate = (sum of H=yes weights)/(sum of all weights) = 1.7000/2.7000 = 17/27.

Likelihood-weighting estimate = 0.6296

3. Compare with the exact 0.5367. Rejection sampling gives 0.4000, an underestimate (absolute error 0.1367); likelihood weighting gives 0.6296, an overestimate (absolute error 0.0930). These are finite-sample estimates from different, very small sample sets, not exact inference results. Increasing the number of independent samples reduces typical sampling error and makes both estimators converge toward the exact answer; no particular finite run is guaranteed to be closer.

Exercise 5(c) — Notebook summary

The implementation and saved execution outputs are in **assignment_bn.ipynb**.

1. Model construction

The notebook uses **DiscreteBayesianNetwork** and **TabularCPD**. Nodes are named heatwave, overload, powercut and cancel, and every node has explicit yes/no state names. The arrows and CPDs exactly match Exercise 3, including powercut's parent order [heatwave, overload].

model.check_model() prints True

2. Exact VariableElimination

Query: heatwave; evidence: {cancel: yes}.
The output is P(heatwave=yes | cancel=yes) = **0.5367** and P(heatwave=no | cancel=yes) = **0.4633**, matching the hand normalization in Exercise 4(1).

3. Rejection sampling

The call uses BayesianModelSampling(model).rejection_sample with evidence=[State("cancel", "yes")], size=10000, seed=42 and show_progress=False. It returns 10,000 retained rows satisfying cancel=yes; **5430** have heatwave=yes.

Sampling estimate = 5430/10000 = 0.5430; absolute error = 0.0063

4. Two-sentence comparison

The exact probability is 0.5367, whereas the rejection-sampling estimate is 0.5430, an absolute difference of 0.0063. This difference is due to finite random sampling; increasing the number of retained samples generally reduces sampling error and makes the estimate approach the exact value.

Verification and interpretation notes

The notebook independently enumerates all 16 joint states using exact fractions. Assertions check the Exercise 3 joint event, both Exercise 3 posteriors, all three Exercise 4 posteriors, and the Exercise 1, 2 and 5(a),(b) arithmetic; overlapping inference answers also agree with pgmpy within 10^{-12}.

Sampling wording: "10,000 samples" could mean prior proposals, but the assignment's explicit size=10000 call returns 10,000 evidence-consistent samples. This solution follows that call; the original number of rejected proposals is not claimed.

Notation: the extracted text shows "24 − 1" in Exercise 3(3); the intended expression is $2^4 - 1 = 15$, consistent with the stated 15 numbers. No ambiguity affects the supplied network probabilities.

Presentation: this PDF is typeset, not handwritten in the original boxes; review instructor formatting requirements before submission. Nothing has been submitted.

Sources: provided CS3081 assignment (all eight pages); official pgmpy sampling, CPD and network documentation (linked in notebook). Package versions are saved in the notebook.